



Bachelor Thesis

Density Functional Theory oriented Optimization of Molecular Geometries using Variational Quantum Eigensolvers

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Abstract

In this thesis, we explore an advanced approach for molecular geometry optimization leveraging the Variational Quantum Eigensolver (VQE). Building on existing quantum-classical hybrid algorithms, the present approach extends applicability to a diverse set of molecules and introduces a novel initialization strategy based on Density Functional Theory (DFT). We hypothesize that DFT-derived initial states provide a closer approximation to the ground state compared to the traditional Hartree-Fock guesses, thereby accelerating convergence and improving optimization efficiency. We evaluate the performance of the proposed method in terms of accuracy, convergence rate, and computational cost. Our findings highlight the potential of integrating DFT with VQE as a promising pathway for enhancing quantum algorithms in computational chemistry.

Keywords

Optimization Algorithm, Variational Quantum Eigensolver (VQE), Density Functional Theory, Quantum Chemistry, Quantum Machine Learning

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Chapter 1

Introduction

The accurate computational modeling of molecular systems is an important task in modern chemistry, with applications for drug discovery, materials science, and catalysis. For decades, this task has been dominated by classical computational methods. However, the computational complexity of solving the electronic structure problem for anything but the smallest molecules poses a significant challenge, often referred to as the curse of dimensionality. As we try modeling larger and more complex systems, these classical methods face growing limitations in both accuracy and computational cost [1].

The fundamental limitations of classical computing for quantum mechanical problems were recognized early on. From the 1980s onwards, scientists like Richard Feynman [2] envisioned calculators not following the laws of classical physics but rather those of quantum physics, resulting in the quantum analogue to the Church-Turing machine [3], what we now call a quantum computer. This paved the way for the development of quantum information theory [4] as the foundation for devices that can exploit the strong parallelism and correlations (entanglement) that occur among physical systems in nature to handle complex problems, thus promising substantial computational speed-up [5, 6].

The innovation of quantum computing introduces a powerful new approach. Founded on the principles of quantum mechanics, quantum computers provide a natural framework for simulating quantum mechanical systems like molecules. Quantum algorithms, in principle, can simulate such systems with a resource cost that scales polynomially with system size, potentially offering an exponential advantage over the best classical algorithms that are known.

In the meantime, several real implementations of such quantum mechanical computing machines have been proposed and are now being developed, with the first experimental demonstrations of exponential advantages [7]. However, current implementations of quantum computers all share a much higher susceptibility to errors than modern classical computers, and the path to building functional fault-tolerant (noise-free) quantum computers [8, 9, 10] is paved with significant challenges.

Today’s quantum hardware belongs to the Noisy Intermediate-Scale Quantum (NISQ) era [11]. These devices, while possessing dozens to hundreds of qubits, are affected by noise and decoherence that prevent the execution of long, fault-tolerant quantum algorithms. This has led to the development of hybrid quantum-classical algorithms designed to leverage the strengths of both approaches while respecting the constraints of current hardware.

In this NISQ scenario, variational quantum algorithms (VQAs) [12] have emerged as promising solutions. These algorithms adopt the concept of training quantum computers in a similar manner to training neural networks, finding the optimum parameters of the model to minimize/maximize some objective function related to that model, typically done via variational methods [13, 14]. This framework has found applications across diverse fields, including optimization, and the emerging discipline of quantum machine

learning (QML) which aims to exploit quantum information processing to enhance the capabilities of intelligent data mining [15, 16]. However, one of the most relevant and foundational applications of VQAs is solving quantum chemistry problems [17, 18, 19, 20], which is the focus of this thesis.

Among the most prominent approaches to quantum chemistry is the Variational Quantum Eigensolver (VQE). The VQE algorithm uses a quantum computer to prepare and measure a parameterized trial wavefunction (the ansatz), while a classical optimizer adjusts the parameters to minimize the expectation value of the molecular Hamiltonian, thus approximating the ground-state energy. In this thesis, we focused on the usage of Variational Quantum Eigensolver to find the ground state of molecular systems. Since VQE is suitable for NISQ hardware, my goal is to explore and optimize this approach for practical use in molecular modeling. We will utilize and examine the Variational Quantum Algorithm approach suggested in the paper [21].

1.1 Outline

This thesis is organized as follows:

Chapter 1 Introduction presents the motivation and context for quantum computing in molecular geometry optimization, outlining the challenges in classical computational chemistry and the potential advantages of quantum algorithms.

Chapter 2 Theoretical Background in Quantum Chemistry provides the essential foundation, introducing the Schrödinger equation, the Born-Oppenheimer approximation, the concept of the Potential Energy Surface (PES), Orbitals, and Classical Computation methods in Quantum Chemistry, which are crucial for geometry optimization.

Chapter 3 Fundamentals of Quantum Computing reviews the basic principles of quantum computation, including qubits, quantum gates, entanglement, and Hamiltonian simulation techniques including fermion-to-qubit transformations.

Chapter 4 Variational Quantum Eigensolver introduces complexity classes and the NISQ era context, then details the VQE algorithm including the variational principle and the geometry-dependent workflow for molecular optimization.

Chapter 5 Results and Discussion presents the findings of molecular geometry optimization experiments for H_3^+ and H_2 systems, comparing Hartree-Fock and DFT initialization approaches and analyzing convergence behavior.

Chapter 6 Conclusions summarizes the key findings, discusses computational limits and implications for quantum computational chemistry, and suggests directions for future research.

Chapter 2

Quantum Chemistry

The accurate prediction of molecular geometries represents a central challenge in computational chemistry. This chapter introduces the quantum mechanical principles that underlie molecular structure determination, beginning with the fundamental equations and progressing to the practical computational methods used to locate energy minima on potential energy surfaces. Understanding these classical approaches is essential for realizing the potential advantages offered by quantum computing algorithms in subsequent chapters.

2.1 The Schrödinger Equation

The Schrödinger equation is the foundational equation of quantum mechanics, describing how the quantum state of a physical system evolves in space and time. It serves as the quantum analogue of Newton’s laws, describing all dynamical and stationary properties of matter [22].

The Hamiltonian, denoted as $\hat{H}$, is a fundamental operator in quantum mechanics and represents the total energy of a quantum system. In the context of molecular systems, it encapsulates the total energy of a molecule, comprising the kinetic energies of all electrons and nuclei and the potential energy arising from their mutual electrostatic interactions (electron–electron repulsion, nucleus–nucleus repulsion, and electron–nucleus attraction). The Hamiltonian is crucial to determine a molecule’s structure, stability, and reactivity.

The central role of the Hamiltonian is defined by the **time-independent Schrödinger equation**:

$$\hat{H}\Psi = E\Psi, \quad (2.1)$$

where Ψ is the wavefunction describing the quantum state of the system and E is the corresponding total energy (eigenvalue). The wavefunction contains all measurable information about the system, while the eigenvalue E corresponds to the quantized energy level associated with that state.

Solving the eigenvalue problem in equation (2.1) for molecular systems is the fundamental challenge of quantum chemistry [23]. The Hamiltonian also describes the system’s time evolution via the **time-dependent Schrödinger equation**:

$$i\hbar \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = \hat{H}\Psi(\mathbf{r}, t), \quad (2.2)$$

which shows that the Hamiltonian acts as the generator of time evolution in quantum mechanics. This can also be written in Dirac notation as:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad (2.3)$$

where $\hat{H}$ is the system Hamiltonian and $|\psi(t)\rangle$ is the state vector at time t ¹.

The formal solution to this equation is given by the unitary time-evolution operator:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle. \quad (2.4)$$

This solution can be written more compactly as:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad (2.5)$$

where $U(t) = e^{-i\hat{H}t/\hbar}$ is the unitary time-evolution operator that satisfies $U^\dagger U = U U^\dagger = I$, preserving the norm of the quantum state during evolution [22].

Simulating this operator on a quantum computer lies at the core of quantum simulation. The goal is to approximate the evolution of an arbitrary initial state $|\psi(0)\rangle$ under the Hamiltonian $\hat{H}$ efficiently. Since the exponential of a general many-body Hamiltonian cannot be evaluated directly, quantum algorithms employ Hamiltonian decomposition and Trotterization techniques to implement this evolution [20].

2.2 The Born-Oppenheimer Approximation & The Electronic Schrödinger Equation

Now, we know that the general Schrödinger equation describes the evolution of a wave function (Ψ) for any quantum particle, such as an electron or atom, over time and space. To apply this in quantum chemistry, a specific form of this equation is used to describe molecular systems: the **electronic Schrödinger equation**.

For molecules, the total Hamiltonian contains contributions from both the nuclei and the electrons. In atomic units, the Hamiltonian can be expressed as:

$$\hat{H} = - \sum_i \frac{\nabla_i^2}{2} - \sum_A \frac{\nabla_A^2}{2M_A} - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i<j} \frac{1}{r_{ij}} + \sum_{A<B} \frac{Z_A Z_B}{R_{AB}}, \quad (2.6)$$

where i, j index electrons and A, B index nuclei, M_A and Z_A denote the mass and charge of nucleus A , r_{ij} is the distance between electrons i and j , R_{AB} is the distance between nuclei A and B , and r_{iA} is the distance between electron i and nucleus A .

The full molecular Schrödinger equation is shown in 2.7. Solving this equation directly is computationally intractable for all systems except the smallest ones (for example, the H_2 molecule). This is because the total wavefunction $\Psi(\mathbf{r}, \mathbf{R})$ depends on both the coordinates of all electrons $\mathbf{r}$ and all nuclei $\mathbf{R}$, leading to an exponential scaling with the number of particles [24].

$$\hat{H}\Psi(\mathbf{r}, \mathbf{R}) = E\Psi(\mathbf{r}, \mathbf{R}) \quad (2.7)$$

The **Born-Oppenheimer approximation** provides a critical simplification to this problem. It exploits the large difference in mass between nuclei and electrons: since nuclei are much heavier, they move significantly slower. Therefore, the nuclei can be considered fixed while solving the electronic problem [20].

Under this approximation, the total molecular wavefunction is separated into electronic and nuclear components:

$$\Psi(\mathbf{r}, \mathbf{R}) = \Psi_{\text{elec}}(\mathbf{r}; \mathbf{R}) \Psi_{\text{nuc}}(\mathbf{R}), \quad (2.8)$$

where $\Psi_{\text{elec}}(\mathbf{r}; \mathbf{R})$ is the electronic wavefunction for fixed nuclei, and $\Psi_{\text{nuc}}(\mathbf{R})$ describes nuclear motion.

Substituting this form into the equation (2.7) and neglecting terms involving nuclear kinetic energy acting on Ψ_{elec} yields the **electronic Schrödinger equation**:

$$\hat{H}_{\text{elec}}(\mathbf{R}) \Psi_{\text{elec}}(\mathbf{r}; \mathbf{R}) = E_{\text{elec}}(\mathbf{R}) \Psi_{\text{elec}}(\mathbf{r}; \mathbf{R}), \quad (2.9)$$

¹The wave function $\Psi(\mathbf{r}, t)$ corresponds to the position representation of the state $|\psi(t)\rangle$.

where $\hat{H}_{\text{elec}}(\mathbf{R})$ is the electronic Hamiltonian:

$$\hat{H}_{\text{elec}}(\mathbf{R}) = - \sum_i \frac{\nabla_i^2}{2} - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i < j} \frac{1}{r_{ij}}. \quad (2.10)$$

Here, $E_{\text{elec}}(\mathbf{R})$ represents the electronic energy, which depends parametrically on the nuclear positions $\mathbf{R}$. For any fixed arrangement of nuclei, solving the electronic Schrödinger equation yields the corresponding electronic energy $E_{\text{elec}}(\mathbf{R})$.

By evaluating the equation (2.9) for many different nuclear configurations, one constructs the **potential energy surface (PES)**. The PES describes how the electronic energy of a molecule changes as the nuclei move and forms the foundation for studying equilibrium geometries, molecular vibrations, and reaction pathways [25]. The following section introduces the PES in detail and explains how stable structures and transition states are located on this surface.

2.3 Potential Energy Surface (PES)

Molecules naturally adopt the geometry that minimizes their energy, and geometry optimization is the process used to identify this most stable structure. The lowest point on the potential energy surface (PES) corresponds to the equilibrium geometry, or ground state, of the molecule.

The potential energy surface is a conceptual and mathematical representation that describes how the energy of a system changes as a function of the positions of its atoms [23]. Each point on the PES corresponds to a unique molecular geometry and its associated potential energy. For a molecule with N atoms, the PES is a $(3N-6)$ -dimensional hypersurface (or $3N-5$ for linear molecules), where the dimensions are the internal degrees of freedom (for example bond lengths, bond angles, dihedral angles) [26]. This shows how the real-world geometry optimization becomes a high-dimensional search problem for big molecules.

In practice, geometry optimization involves navigating this surface to locate stationary points where the energy no longer decreases with small changes in geometry.

$$\frac{\partial E}{\partial R_i} = 0 \quad \forall i, \quad (2.11)$$

where R_i denotes the Cartesian coordinate of nucleus i .

Figure 2.1 illustrates the essential features of a PES using a simplified one-dimensional coordinate (R), which represents the bond length between two atoms.

Stationary points are shown on the left graph. These occur where the slope of the PES is zero, which happens when the first derivative of the energy with respect to R vanishes: $\frac{\partial E}{\partial R} = 0$.

Both the ground state (a minimum) and the transition state (a saddle point) share this property of having zero slope.

On the right graph, the distinction between minima and transition states is shown using the second derivative of the energy, which defines the curvature: $\frac{\partial^2 E}{\partial R^2}$.

At a minimum (ground state), the curvature is positive, meaning the energy increases in all directions away from the equilibrium point. This corresponds to a stable molecular geometry. At a transition state, the curvature is negative along at least one coordinate, meaning the structure is unstable in that direction. This represents the highest-energy configuration along a reaction pathway.

In summary, geometry optimization seeks to find minima on the PES, characterized by a vanishing gradient (zero first derivative) and positive curvature (second derivative > 0). Transition state searches instead to locate saddle points, where the gradient also vanishes but the curvature is negative along at least one coordinate.

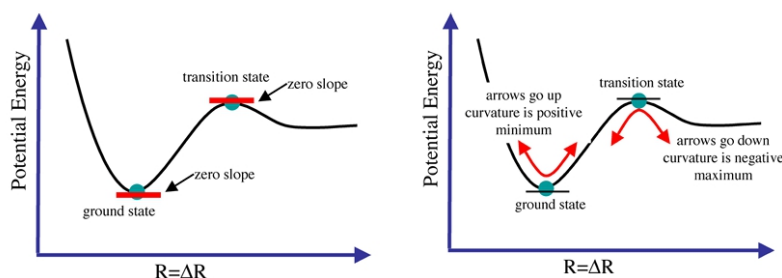


Figure 2.1: Illustration of the potential energy surface (PES). **Left:** ground state and transition state both have a zero slope (first derivative = 0). **Right:** the second derivative describes the curvature, distinguishing minima (positive curvature) from maxima (negative curvature). [27]

The central computational challenge in geometry optimization is the repeated evaluation of the electronic energy $E_{\text{elec}}(\mathbf{R})$ and its derivatives (the gradient) for many different nuclear configurations $\mathbf{R}$. While classical methods like Hartree-Fock (HF) are tractable on classical computers, they often lack the accuracy required for predictive chemistry because they do not fully account for electron correlation [28]. This motivates the exploration of quantum computing algorithms, such as the Variational Quantum Eigensolver (VQE), which in principle can provide a more efficient route to solving the electronic Schrödinger equation and thus constructing the PES [20, 17, 21]. Therefore, we will explore the integration of VQE-generated energies into geometry optimization workflows.

2.4 The Variational Principle

When the exact solution of the Schrödinger equation is not feasible, the Variational Principle provides a powerful method to approximate the ground-state energy of a quantum system [23, 20].

In its general form, the variational principle states that for any normalized trial wave function Ψ_{trial} , the expectation value of the Hamiltonian provides an upper bound to the true ground-state energy E_0 :

$$E_{\text{trial}} = \frac{\langle \Psi_{\text{trial}} | \hat{H} | \Psi_{\text{trial}} \rangle}{\langle \Psi_{\text{trial}} | \Psi_{\text{trial}} \rangle} \geq E_0 \quad (2.12)$$

Equality holds only when Ψ_{trial} is exactly the ground-state wave function Ψ_0 . This principle follows directly from the spectral decomposition of the Hamiltonian operator, which ensures that any wave function can be expressed as a linear combination of eigenstates of $\hat{H}$ [22].

The method can be viewed as an optimization problem: it searches for the set of parameters θ that minimize the energy expectation value,

$$E(\theta) = \langle \Psi(\theta) | \hat{H} | \Psi(\theta) \rangle, \quad (2.13)$$

where $\Psi(\theta)$ is a parameterized trial wave function. The goal is to find

$$E_0 = \min_{\theta} E(\theta). \quad (2.14)$$

In classical computational chemistry, this optimization is performed using analytical or numerical techniques over basis functions, as in the Hartree-Fock method. In quantum computing, the same principle underlies hybrid algorithms such as the *Variational Quantum Eigensolver* (VQE) [13, 12, 14], where $\Psi(\theta)$ is prepared by a parameterized quantum circuit and the expectation value is evaluated using a quantum processor. (See Chapter 3 for the introduction to these topics.)

2.5 Spins and Orbitals

To describe the electronic structure of atoms and molecules, each electron is represented by a wave function called a *spin orbital* [23, 20]. A spin orbital, $\chi(\mathbf{x})$, depends on both the spatial coordinates $\mathbf{r}$ and the spin coordinate ω of an electron, such that $\mathbf{x} = \{\mathbf{r}, \omega\}$. It can be separated into a spatial part, $\phi(\mathbf{r})$, which defines the probability distribution of finding an electron in space, and a spin part, $\sigma(\omega)$, representing the two possible spin states of an electron: spin-up (α) or spin-down (β).

$$\chi(\mathbf{x}) = \phi(\mathbf{r}) \sigma(\omega) \quad (2.15)$$

The spatial orbitals $\phi(\mathbf{r})$ describe how electrons are distributed around the nuclei. For molecules, these orbitals are typically expressed as linear combinations of atomic orbitals (LCAO) [23]:

$$\phi_i(\mathbf{r}) = \sum_{\mu} c_{\mu i} \phi_{\mu}(\mathbf{r}), \quad (2.16)$$

where ϕ_{μ} are atomic orbitals and $c_{\mu i}$ are coefficients determined through energy minimization methods such as the Hartree–Fock approach.

Each electron occupies a distinct spin orbital, and the total many-electron wave function is constructed to satisfy the Pauli exclusion principle, ensuring antisymmetry under particle exchange. This antisymmetric property is achieved mathematically using a *Slater determinant*, which proves that no two electrons can occupy the same spin orbital.

Constructing accurate spin orbitals is essential for both classical and quantum computational methods. For instance, the hydrogen atom has a single 1s orbital, while oxygen possesses 1s, 2s, and 2p atomic orbitals. Each spatial orbital can host two electrons of opposite spin (spin-up α and spin-down β), leading to twice as many spin orbitals as spatial orbitals. This relationship demonstrated in Figures 2.2 and 2.3 forms the basis for representing the electronic structure of atoms and molecules.



Figure 2.2: Hydrogen atom with a single 1s orbital containing two electrons of opposite spin. The hydrogen atom has only one spatial orbital, giving rise to two spin orbitals corresponding to spin-up (α) and spin-down (β) states.

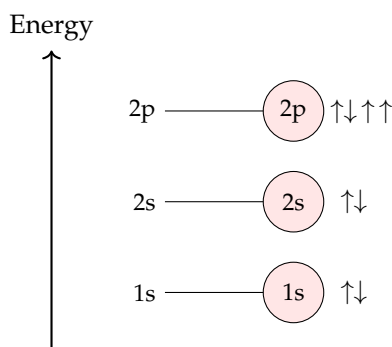


Figure 2.3: Energy-level diagram for oxygen showing occupied 1s, 2s, and 2p orbitals. Each orbital gives rise to two spin orbitals corresponding to the two possible spin projections (α and β). The 2p subshell contains four unpaired electrons following Hund’s rule.

In the context of quantum algorithms such as the VQE, the molecular spin orbitals are encoded onto qubits using fermion-to-qubit transformations, most commonly the Jordan–Wigner or Bravyi–Kitaev mappings [29, 20]. These transformations (discussed in

Section 3.6) translate the second-quantized electronic Hamiltonian, expressed in terms of creation and annihilation operators, into a qubit Hamiltonian composed of tensor products of Pauli operators. This enables quantum computers to represent and manipulate molecular wave functions directly, allowing for the efficient estimation of electronic energies.

2.6 Classical Computation Methods

In quantum chemistry, solving the electronic Schrödinger equation exactly for many-electron systems is infeasible due to the exponential scaling of the Hilbert space with the number of electrons [20, 17]. Therefore, a hierarchy of approximate methods has been developed to obtain tractable yet accurate solutions. Two fundamental approaches are the *Hartree–Fock (HF)* method and the *Configuration Interaction (CI)* method. Both are rooted in the variational principle (see Section 2.4) but differ in how they approximate the many-electron wavefunction.

2.6.1 The Hartree–Fock Method

The Hartree–Fock (HF) method provides a mean-field approximation to the electronic structure of atoms and molecules [23]. In this framework, each electron moves in the average electrostatic field produced by all other electrons, neglecting instantaneous electron-electron correlations.

The total electronic wavefunction is approximated by a single *Slater determinant*, which ensures that the overall wavefunction is antisymmetric with respect to exchange of any two electrons, in accordance with the Pauli exclusion principle:

$$\Psi_{\text{HF}}(\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(\mathbf{x}_1) & \chi_2(\mathbf{x}_1) & \cdots & \chi_N(\mathbf{x}_1) \\ \chi_1(\mathbf{x}_2) & \chi_2(\mathbf{x}_2) & \cdots & \chi_N(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_1(\mathbf{x}_N) & \chi_2(\mathbf{x}_N) & \cdots & \chi_N(\mathbf{x}_N) \end{vmatrix}, \quad (2.17)$$

where $\chi_i(\mathbf{x})$ denotes a spin orbital as introduced in Section 2.5.

The energy of this trial wavefunction is minimized with respect to variations in the spin orbitals under the constraint that they remain orthonormal. This leads to the *Hartree–Fock equations*:

$$\hat{F}\chi_i = \varepsilon_i \chi_i, \quad (2.18)$$

where $\hat{F}$ is the Fock operator, which includes the one-electron kinetic and potential energy terms as well as the average electron-electron interaction through Coulomb and exchange operators. The resulting eigenvalues ε_i correspond to orbital energies, and the lowest total energy obtained defines the Hartree–Fock approximation to the ground-state energy:

$$E_{\text{HF}} = \langle \Psi_{\text{HF}} | \hat{H} | \Psi_{\text{HF}} \rangle. \quad (2.19)$$

Although the HF method captures exchange effects due to the antisymmetry of the wavefunction, it neglects *electron correlation* (the dynamic coupling between electron motions). Consequently, HF energies are systematically higher than the true ground-state energies.

2.6.2 The Configuration Interaction Method

To improve upon the limitations of Hartree–Fock, post-HF methods explicitly include electron correlation. One of the most fundamental approaches is the *Configuration Interaction (CI)* method.

In CI, the exact many-electron wavefunction is expressed as a linear combination of multiple Slater determinants (configurations):

$$\Psi_{\text{CI}} = c_0 \Psi_{\text{HF}} + \sum_i c_i \Psi_i^{(1)} + \sum_{i < j} c_{ij} \Psi_{ij}^{(2)} + \dots, \quad (2.20)$$

where Ψ_{HF} is the reference (ground-state) determinant, and $\Psi_i^{(1)}$, $\Psi_{ij}^{(2)}$, etc. represent excited determinants generated by promoting one or more electrons from occupied to virtual orbitals. The coefficients $\{c_k\}$ are determined variationally to minimize the total energy.

When all possible excitations are included, the method is known as *Full Configuration Interaction (Full CI)*, which provides an exact solution within the chosen basis set. However, the computational cost of Full CI scales factorially with the number of electrons and orbitals, making it impractical for anything beyond very small molecules. Therefore, truncated versions such as *CISD* (Configuration Interaction with Single and Double excitations) are often used in practice.

While CI methods significantly improve accuracy by incorporating electron correlation, their steep computational scaling motivates the exploration of more efficient alternatives such as VQE, which aim to achieve similar accuracy with polynomial scaling on quantum hardware [28, 20].

2.6.3 Density Functional Theory Method

Density Functional Theory (DFT) provides an alternative approach to electronic structure calculation that, unlike Hartree–Fock, explicitly accounts for electron correlation while maintaining favorable computational scaling [20]. Rather than working with the complex many-electron wavefunction, DFT operates on the electron density $n(\mathbf{r})$, a simpler function of only three spatial coordinates.

The theoretical foundation of DFT rests on the Hohenberg–Kohn theorems [30]. The first theorem establishes that the ground-state electron density $n_0(\mathbf{r})$ uniquely determines all electronic properties of the system. The second theorem provides a variational principle: the true ground-state density minimizes the energy functional $E[n]$, where E depends functionally on the electron density $n(\mathbf{r})$.

The Kohn–Sham approach [31] makes DFT practical by replacing the complex interacting electron system with a simpler system of non-interacting electrons that has the same electron density. The total energy is expressed as:

$$E[n] = T_s[n] + E_{\text{ext}}[n] + E_H[n] + E_{\text{xc}}[n], \quad (2.21)$$

where $T_s[n]$ is the kinetic energy of non-interacting electrons, $E_{\text{ext}}[n]$ is the electron-nucleus attraction, $E_H[n]$ is the classical electron-electron repulsion and $E_{\text{xc}}[n]$ is the exchange-correlation energy (containing all quantum effects).

The Kohn–Sham equations are solved self-consistently:

$$\left[-\frac{1}{2} \nabla^2 + v_{\text{eff}}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad (2.22)$$

where $\phi_i(\mathbf{r})$ are the Kohn–Sham orbitals (single-electron wavefunctions), ε_i are the corresponding orbital energies, $-\frac{1}{2} \nabla^2$ is the kinetic energy operator, and $v_{\text{eff}}(\mathbf{r})$ is the effective potential that combines nuclear attraction (electron-nucleus interactions), electron repulsion (classical Coulomb interactions between electrons), and exchange-correlation effects (quantum mechanical interactions including electron exchange and correlation).

The main challenge in DFT is that the exact form of the exchange-correlation functional $E_{\text{xc}}[n]$ is unknown. Different approximations to this functional lead to different levels of accuracy:

Common approximations include:

- **LDA (Local Density Approximation):** Uses the exchange-correlation energy of a uniform electron gas;
- **GGA (Generalized Gradient Approximation):** Includes dependence on the density gradient (for example, PBE [32], BLYP);
- **Hybrid functionals:** Mix Hartree–Fock exchange with DFT correlation (for example, B3LYP [33]);

DFT typically scales as $\mathcal{O}(N^3)$ with system size, making it more efficient than correlated wavefunction methods while generally providing better accuracy than Hartree–Fock. This favorable balance between computational cost and accuracy has made DFT one of the most widely used methods in computational chemistry and materials science [34].

Chapter 3

Quantum Computing

As established in Chapter 1, quantum computing represents a shift from classical computation that utilizes the fundamental principles of quantum mechanics to solve problems intractable for classical computers. While classical computation relies on bits that take binary values (0 or 1), quantum computation is based on qubits, which can exist in superpositions of both states simultaneously. This unique property, combined with entanglement and quantum interference, enables quantum algorithms to explore large computational spaces more efficiently than their classical counterparts [35].

The theoretical foundation for quantum computing, laid out by pioneers like Feynman [2] and Deutsch [3], has evolved from abstract concept to experimental reality in recent decades. From the early theoretical models to today’s Noisy Intermediate-Scale Quantum (NISQ) devices [36, 11], the field has developed a rich framework of quantum algorithms and computational models that can achieve exponential speedups for specific problems, most notably quantum simulation [37] and factoring [6]. These unique quantum capabilities extend far beyond computation to fields such as cryptography [38, 39], optimization, and materials science.

The following sections introduce the fundamental concepts of quantum computing needed to understand our methodology, including qubits, quantum gates and quantum circuits. These concepts provide the necessary foundation for understanding the Variational Quantum Eigensolver algorithm and its application to molecular geometry optimization in Chapter 4.

3.1 Quantum Systems

In quantum computing, we typically consider finite-dimensional quantum systems defined by Hilbert spaces $\mathcal{H} = \mathbb{C}^n$, where $n \in \mathbb{N}$. A pure state of such a system is represented by a unit vector $|\psi\rangle \in \mathcal{H}$ with $\|\psi\| = 1$, where the normalization ensures that probabilities sum to one in any measurement.

In contrast, the quantum chemistry problems introduced in Chapter 2 involve infinite-dimensional Hilbert spaces, as molecules are described by continuous parameters representing the positions of electrons and nuclei in three-dimensional space. The wavefunction in the equation (2.1) corresponds to a quantum state in this infinite-dimensional space. The goal of quantum computational chemistry is to use finite-dimensional quantum computers to approximate or compute properties of these complex continuous systems [17, 20].

Throughout this chapter, we employ Dirac notation: $|\psi\rangle$ denotes a quantum state vector, $\langle\psi|$ its dual, and $\langle\psi|\phi\rangle$ represents the inner product. The dagger ($\dagger$) indicates the Hermitian conjugate of an operator or matrix.

3.2 Qubits and Superposition

A qubit, or quantum bit, is the fundamental unit of quantum information, represented by a two-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^2$. The computational basis consists of the orthonormal states:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (3.1)$$

as defined in [35].

Unlike a classical bit, which is either in the state $|0\rangle$ or $|1\rangle$, a qubit can be in a linear combination of both:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (3.2)$$

where α and β are complex probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$.

This superposition property allows qubits to encode a continuum of states on the surface of the *Bloch sphere* (Figure 3.1), a geometric representation of the qubit's state space [35]. The north and south poles correspond to the basis states $|0\rangle$ and $|1\rangle$, while any point on the sphere's surface represents a valid superposition. States on the surface are called *pure states*, as they represent states of complete knowledge about the quantum system. In contrast, points inside the sphere represent *mixed states*, which are probabilistic mixtures of pure states and describe situations of incomplete knowledge.

Using spherical coordinates, a pure state can be parameterized as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle, \quad (3.3)$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ determine the state's orientation.

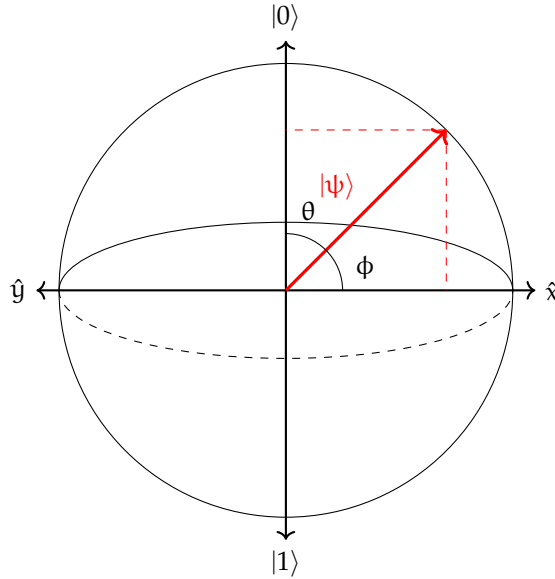


Figure 3.1: The Bloch sphere representation of a single qubit. The state $|\psi\rangle$ (red arrow) is parameterized by angles θ and ϕ , with the poles representing the computational basis states $|0\rangle$ and $|1\rangle$. The equatorial plane encodes phase information.

Measurement in the computational basis collapses the qubit into either $|0\rangle$ or $|1\rangle$ with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively. This probabilistic nature of measurement, even if performed on pure states, distinguishes quantum computation from deterministic classical computation [35].

3.3 Composite Systems and Entanglement

Composite quantum systems are constructed using the tensor product of Hilbert spaces. A typical system in quantum computation consists of n qubits, with the joint Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ [35].

The tensor product $\otimes$ combines individual quantum states. For example, the joint state of two qubits, one in state $|0\rangle$ and the other in $|1\rangle$, is written as $|0\rangle \otimes |1\rangle$, often abbreviated as $|01\rangle$. The computational basis of a two-qubit system comprises the four classical bitstring states: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Since any linear superposition of quantum states is still a good quantum state, a general two-qubit state is expressed as:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle, \quad (3.4)$$

where each α_{ij} is a complex amplitude, and the normalization condition

$$\sum_{x \in \{0,1\}^2} |\alpha_x|^2 = 1$$

ensures valid probabilities upon measurement.

When multiple qubits interact, they can form composite states that cannot be expressed as simple products of individual qubit states. Such states are called *entangled*. Entanglement is a unique quantum phenomenon that enables qubits to exhibit correlations beyond what is possible classically [39].

For example, the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (3.5)$$

describes two qubits whose measurement outcomes are perfectly correlated when measured along the same axis of the Bloch sphere.

Formally, a pure state of two qubits is entangled if it cannot be factorized:

$$|\Psi\rangle \neq |\phi\rangle \otimes |\eta\rangle \quad \forall |\phi\rangle, |\eta\rangle \in \mathbb{C}^2. \quad (3.6)$$

One can verify that the Bell state in (3.5) is entangled according to this criterion.

3.4 Dynamics: Quantum Gates

The evolution of quantum systems via unitary operators, as introduced in Chapter 2, applies directly to quantum computation. For finite-dimensional systems of n -qubits, the evolution is governed by a unitary operator:

$$\begin{aligned} U : (\mathbb{C}^2)^{\otimes n} &\rightarrow (\mathbb{C}^2)^{\otimes n}, \\ |\psi\rangle &\rightarrow U|\psi\rangle, \end{aligned} \quad (3.7)$$

which can be represented as a $2^n \times 2^n$ unitary matrix satisfying $U^\dagger U = U U^\dagger = I$ [35].

In analogy to classical logic gates (OR, NOT, AND, etc.), these unitary operators process information encoded in quantum systems and are therefore called **quantum gates**. The following sections provide examples of fundamental 1-qubit and 2-qubit gates commonly used in quantum circuits.

3.4.1 Pauli Operators and Single-Qubit Gates

The fundamental building blocks of quantum gates are the Pauli operators, which form a basis for single-qubit operations:

- **Pauli-X gate (X):** The quantum equivalent of the classical NOT gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle. \quad (3.8)$$

- **Pauli-Y gate (Y):** Performs a rotation around the y-axis of the Bloch sphere

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle. \quad (3.9)$$

- **Pauli-Z gate (Z):** Performs a phase flip (rotation around the z-axis)

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle. \quad (3.10)$$

- **Identity operator (I):** Leaves the qubit unchanged

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (3.11)$$

These Pauli operators satisfy the commutation relations $[X, Y] = 2iZ$ (cyclic permutations) and form a complete basis for 2×2 Hermitian matrices. Any single-qubit gate can be expressed as a linear combination of Pauli operators.

3.4.2 Other Important Single-Qubit Gates

Beyond the Pauli gates, several other single-qubit gates are essential for quantum computation:

- **Hadamard gate (H):** Creates superposition states

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (3.12)$$

- **Phase gates (S, T):** Special cases of Z-rotations

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}. \quad (3.13)$$

- **Rotation gates:** Parameterized rotations around Bloch sphere axes

$$R_x(\theta) = e^{-i\theta X/2}, \quad R_y(\theta) = e^{-i\theta Y/2}, \quad R_z(\theta) = e^{-i\theta Z/2}. \quad (3.14)$$

3.4.3 Multi-Qubit Gates and Entanglement

While single-qubit gates manipulate individual qubits, multi-qubit gates can create correlations and entanglement between qubits:

- **CNOT gate (controlled-NOT):** A two-qubit gate that flips the target qubit if the control qubit is $|1\rangle$

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|10\rangle = |11\rangle. \quad (3.15)$$

- **Controlled-phase gates:** Apply phase shifts conditioned on control qubits

The combination of single-qubit gates with entangling two-qubit gates like CNOT forms a universal set for quantum computation, enabling the implementation of any unitary operation to arbitrary precision.

In summary, quantum gates manipulate qubit states through unitary transformations. Single-qubit gates, such as the Pauli-X, Y, and Z operators, perform rotations around the Bloch sphere axes, while two-qubit gates such as the CNOT gate can generate entanglement. A universal set of quantum gates can approximate any unitary operation to arbitrary precision. Common examples of universal gate sets include $\{H, T, \text{CNOT}\}$ [35], where H is the Hadamard gate, T is the $\pi/8$ gate, and CNOT is the controlled-NOT gate.

3.5 Quantum Circuits

Quantum circuits are the quantum analog of classical logic circuits, representing computations as sequences of gate operations acting on qubits. A general n -qubit circuit applies a unitary transformation U to an initial state $|\psi_0\rangle$, producing

$$|\psi_{\text{out}}\rangle = U |\psi_0\rangle, \quad |\psi_0\rangle \in (\mathbb{C}^2)^{\otimes n}. \quad (3.16)$$

At the end of the computation, measurements produce classical outcomes that can be statistically analyzed.

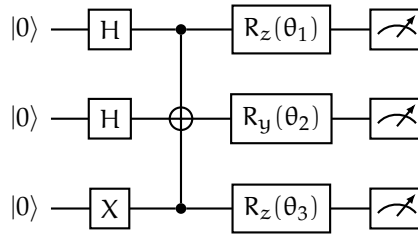


Figure 3.2: A three-qubit quantum circuit demonstrating common quantum operations. The circuit includes Hadamard gates (H) for superposition creation, Pauli-X gate for bit-flip, controlled-NOT (CNOT) gates for entanglement, parameterized rotation gates (R_y , R_z) for state preparation, and final measurement operations.

Quantum circuits can be implemented on various hardware platforms, including superconducting qubits, trapped ions, and photonic systems [7, 19]. While these devices are still limited by decoherence and noise, hybrid algorithms such as the VQE leverage both classical and quantum computation to mitigate these challenges [12, 21].

3.6 Simulation of the Hamiltonian

As discussed in Section 2.1, the time evolution of a quantum system is governed by the time-dependent Schrödinger equation, whose formal solution involves the unitary time-evolution operator $e^{-i\hat{H}t/\hbar}$. For molecular systems, implementing this operator on a quantum computer requires mapping the electronic Hamiltonian to a form suitable for quantum computation.

The Hamiltonian of a molecular system can be expressed in the second-quantized form as:

$$\hat{H} = \sum_{p,q} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{p,q,r,s} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s, \quad (3.17)$$

where the coefficients h_{pq} and h_{pqrs} correspond to the one- and two-electron integrals over the chosen basis set, and $a_p^\dagger$, a_q are fermionic creation and annihilation operators satisfying the canonical anticommutation relations.

Once expressed in terms of these operators, $\hat{H}$ can be mapped to a qubit Hamiltonian $\hat{H}_q$ through fermion-to-qubit transformations such as the Jordan–Wigner or Bravyi–Kitaev mappings (see Sections 3.6.1 and 3.6.2). The resulting Hamiltonian can be written as a sum of tensor products of Pauli matrices:

$$\hat{H}_q = \sum_j c_j P_j, \quad (3.18)$$

where $P_j \in \{I, X, Y, Z\}^{\otimes n}$ are Pauli strings acting on n qubits and c_j are real coefficients.

The time evolution operator can then be approximated as:

$$e^{-i\hat{H}_q t} \approx \left(\prod_j e^{-ic_j P_j t/m} \right)^m, \quad (3.19)$$

where m is the number of Trotter steps. Increasing m improves accuracy at the cost of a deeper quantum circuit.

In practice, many chemistry algorithms such as the Variational Quantum Eigensolver (VQE) do not simulate full time evolution but instead use measurements of the expectation value

$$E(\theta) = \langle \Psi(\theta) | \hat{H}_q | \Psi(\theta) \rangle, \quad (3.20)$$

to approximate the ground-state energy of $\hat{H}$. Nevertheless, the ability to approximate $e^{-i\hat{H}t}$ efficiently is fundamental to both variational and dynamical quantum simulation methods.

3.6.1 Jordan-Wigner Transformation

The Jordan-Wigner (JW) transformation provides a direct and conceptually simple mapping from fermionic operators to qubit operators. It encodes the occupation of each spin orbital into the computational basis states of qubits, where $|0\rangle$ represents an unoccupied orbital and $|1\rangle$ an occupied one [29].

Under this mapping, the fermionic creation and annihilation operators are expressed in terms of Pauli matrices as:

$$a_j^\dagger = \frac{1}{2} (X_j - iY_j) \prod_{k=0}^{j-1} Z_k, \quad (3.21)$$

$$a_j = \frac{1}{2} (X_j + iY_j) \prod_{k=0}^{j-1} Z_k, \quad (3.22)$$

where X_j, Y_j, Z_j are the Pauli matrices acting on the j -th qubit.

The string of Z operators, known as the *parity string*, ensures that the correct fermionic antisymmetry is preserved by introducing a phase factor corresponding to the number of occupied orbitals preceding index j . Although straightforward to implement, the Jordan-Wigner transformation introduces non-locality, as operators acting on higher-index orbitals affect all preceding qubits. This can lead to increased circuit depth for large systems.

3.6.2 Bravyi-Kitaev Transformation

The Bravyi-Kitaev (BK) transformation offers an alternative mapping that optimizes the trade-off between operator locality and parity encoding. Unlike the Jordan-Wigner approach, which relies entirely on a linear ordering of orbitals, the Bravyi-Kitaev scheme distributes parity and occupation information more evenly across qubits using a binary tree structure [29].

This mapping can be defined recursively, resulting in a logarithmic scaling of parity information propagation. In practical terms, the BK transformation reduces the number of qubits each operator acts upon, improving computational efficiency for quantum simulations:

$$a_j^\dagger \rightarrow \frac{1}{2} (X_j - iY_j) \otimes U_j, \quad a_j \rightarrow \frac{1}{2} (X_j + iY_j) \otimes U_j, \quad (3.23)$$

where U_j encodes the specific parity and occupation update rules determined by the transformation matrix of the Bravyi-Kitaev scheme.

In general, the BK mapping results in shorter Pauli strings than the JW mapping, which can be advantageous for reducing gate depth and decoherence in quantum circuits. However, its implementation is more complex, requiring precomputed transformation matrices and additional logic to track parity relationships [20].

Chapter 4

Variational Quantum Eigensolver

4.1 Complexity Classes and the NISQ Era

Every computational problem can be categorized according to its time and space (memory) complexity, defining a hierarchy of complexity classes. In the quantum context, these classes help us understand what can be solved efficiently by different computational models [40].

For example:

- **BPP** (Bounded-error Probabilistic Polynomial time): the class of problems efficiently solvable by a classical randomized algorithm that runs in polynomial time with bounded error. Examples include:
 - **Polynomial Identity Testing**: Determining if two algebraic expressions are identical. [41]
 - **Miller-Rabin primality test**: probabilistic algorithm that determines whether a number is prime by repeatedly checking if it behaves like a prime under modular exponentiation. [42]
- **BQP** (Bounded-error Quantum Polynomial time): the class of problems efficiently solvable by a fault-tolerant quantum computer operating in polynomial time with bounded error. Notable examples believed to be in BQP but not in BPP include:
 - **Integer factorization** (as demonstrated by Shor’s algorithm). [6]
 - **Discrete logarithm problem**: Finding an integer x such that $g^x \equiv h \pmod{p}$ for given g , h , and prime p , a task that is computationally hard for classical computers but efficiently solvable using Shor’s quantum algorithm. [6]
 - **Simulation of quantum systems**, including molecular Hamiltonians as explored in this thesis.

Recent work by Chen *et al.* [43] provided a formal complexity-theoretic framework for analyzing **NISQ** (Noisy Intermediate-Scale Quantum) devices, a term originally coined by Preskill [36] to describe noisy, intermediate-scale quantum devices that combine quantum and classical resources.

Under their formulation, the proposed inclusion relations are:

$$\text{BPP} \subseteq \text{NISQ} \subseteq \text{BQP}.$$

A **NISQ algorithm** in this framework is characterized as a hybrid quantum-classical procedure, where a classical computer orchestrates the operation of a noisy quantum processor. The process typically involves:

1. preparing an initial (possibly noisy) quantum state,

2. applying sequences of imperfect quantum gates, and
3. performing projective measurements to extract classical information.

Because every quantum operation in this regime is subject to noise and decoherence, a problem is considered to lie within the NISQ regime if it can be solved with high probability and in polynomial time under such constraints. This characterization motivates the development of variational hybrid algorithms, such as the **Variational Quantum Eigensolver (VQE)**, which use classical optimization loops to compensate for quantum noise and limited circuit depth.

4.2 The Variational Quantum Eigensolver

The **Variational Quantum Eigensolver (VQE)** is a hybrid quantum-classical algorithm designed for the Noisy Intermediate-Scale Quantum (NISQ) era. It combines the variational principle of quantum mechanics with classical optimization techniques to estimate the ground-state energy of a Hamiltonian, making it one of the most promising methods for quantum chemistry and materials science applications [14] [13].

4.2.1 Variational Principle and Algorithm Overview

The VQE is grounded in the **variational principle**, which states that the expectation value of a Hamiltonian with respect to any normalized trial wavefunction provides an upper bound to the true ground-state energy:

$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle, \quad (4.1)$$

where $|\psi(\theta)\rangle$ is a parameterized quantum state prepared by a quantum circuit. The objective is to find the parameter vector θ that minimizes this energy expectation value:

$$E_0 = \min_{\theta} E(\theta). \quad (4.2)$$

The algorithm consists of two synergistic components:

1. A **quantum processor** that prepares the parameterized state $|\psi(\theta)\rangle$ and evaluates the expectation value of $\hat{H}$ via repeated measurements.
2. A **classical optimizer** that updates the circuit parameters θ based on the measured energy to minimize $E(\theta)$ iteratively.

This hybrid feedback loop is illustrated in Figure 4.1.

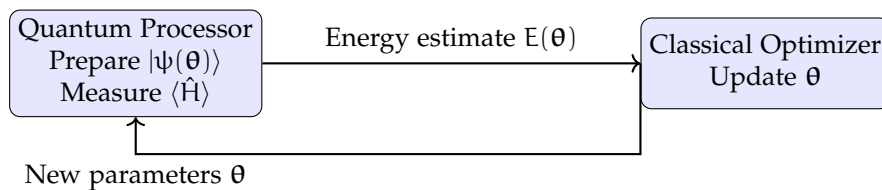


Figure 4.1: Schematic workflow of the Variational Quantum Eigensolver (VQE). The quantum device evaluates the energy expectation value for a parameterized state, while a classical optimizer iteratively updates the parameters until convergence.

Compared to classical electronic structure methods such as Hartree-Fock or Configuration Interaction, VQE can in principle capture strong electron correlation effects more efficiently by exploiting the exponentially large Hilbert space accessible to quantum circuits. However, its performance strongly depends on the expressibility of the chosen ansatz (the parameterized circuit structure) and the optimization strategy used to navigate a potentially noisy, non-convex energy landscape.

4.2.2 Geometry-Dependent Variational Quantum Eigensolver (VQE) Workflow

In the framework proposed by Delgado *et al.* [21], the Variational Quantum Eigensolver (VQE) is extended to perform not only electronic structure calculations but also **molecular geometry optimization**. This hybrid quantum-classical method jointly minimizes the electronic energy with respect to both the quantum circuit parameters θ and the nuclear coordinates $\mathbf{X}$:¹

$$E(\theta, \mathbf{X}) = \langle \Psi(\theta) | \hat{H}_{\text{elec}}(\mathbf{X}) | \Psi(\theta) \rangle, \quad (4.3)$$

where $\hat{H}_{\text{elec}}(\mathbf{X})$ is the electronic Hamiltonian parametrically dependent on the nuclear configuration $\mathbf{X}$, and $|\Psi(\theta)\rangle$ is the parameterized trial wavefunction generated by a quantum circuit.

The goal is to minimize the total energy jointly over both sets of parameters:

$$(\theta^*, \mathbf{X}^*) = \arg \min_{\theta, \mathbf{X}} E(\theta, \mathbf{X}),$$

thus obtaining both the electronic ground state and the equilibrium molecular geometry within a unified variational framework.

Gradient Evaluation and Optimization

Gradients with respect to circuit parameters θ are computed analytically using the **parameter-shift rule** [44]:

$$\frac{\partial E}{\partial \theta_i} = \frac{1}{2} [E(\theta + \frac{\pi}{2} \mathbf{e}_i, \mathbf{X}) - E(\theta - \frac{\pi}{2} \mathbf{e}_i, \mathbf{X})],$$

where $\mathbf{e}_i$ is a unit vector in parameter space. This rule provides an exact, hardware-efficient gradient evaluation without requiring finite-difference approximations.

Gradients with respect to nuclear coordinates $\mathbf{X}$ correspond to the Hellmann-Feynman forces:

$$\frac{\partial E}{\partial X_\alpha} = \langle \Psi(\theta) | \frac{\partial \hat{H}_{\text{elec}}(\mathbf{X})}{\partial X_\alpha} | \Psi(\theta) \rangle.$$

These forces are obtained using classical derivatives of molecular integrals [45] combined with quantum expectation values, allowing efficient computation of nuclear gradients.

Hybrid Optimization Workflow

The VQE geometry optimization follows a hybrid quantum-classical iterative loop (Figure 4.2), which can be summarized as follows:

1. **Initialization:** Begin with an initial molecular geometry $\mathbf{X}^0$, typically obtained from experimental data or a classical pre-optimization (for example Hartree-Fock or DFT). Construct the corresponding electronic Hamiltonian $\hat{H}(\mathbf{X}^0)$ in a chosen basis.
2. **Quantum State Preparation:** A parameterized quantum circuit $\hat{U}(\theta)$ prepares the trial wavefunction $|\Psi(\theta)\rangle = \hat{U}(\theta) |\text{HF}\rangle$, where $|\text{HF}\rangle$ is the Hartree-Fock reference state. The ansatz structure (for example UCC or hardware-efficient) balances expressibility and hardware cost.
3. **Cost Function Evaluation:** The energy expectation value is evaluated as

$$g(\theta, \mathbf{X}) = \langle \Psi(\theta) | \hat{H}(\mathbf{X}) | \Psi(\theta) \rangle,$$

where $\hat{H}(\mathbf{X})$ is decomposed into Pauli operators measured on the quantum device.

¹In the notation of Delgado *et al.* [21], nuclear coordinates are denoted by $\mathbf{X}$, equivalent to $\mathbf{R}$ used elsewhere in this thesis.

4. **Joint Optimization:** Both θ and $\mathbf{X}$ are updated to minimize $g(\theta, \mathbf{X})$ using gradient information:

$$\nabla_{\theta} g(\theta, \mathbf{X}), \quad \nabla_{\mathbf{X}} g(\theta, \mathbf{X}) = \langle \Psi(\theta) | \frac{\partial \hat{H}(\mathbf{X})}{\partial \mathbf{X}} | \Psi(\theta) \rangle.$$

These guide a classical optimizer (for example Gradient Descent, Adam).

5. **Update and Convergence:** Updates $(\theta, \mathbf{X})$ into $(\theta', \mathbf{X}')$ and checks convergence:

$$|E_{k+1} - E_k| < \epsilon, \quad \|\mathbf{X}_{k+1} - \mathbf{X}_k\| < \delta.$$

The process continues until both energy and geometry converge below set thresholds.

6. **Output:** The final optimized quantities $(\theta^*, \mathbf{X}^*)$ yield:

$$E^* = g(\theta^*, \mathbf{X}^*),$$

corresponding to the electronic ground-state energy and equilibrium geometry.

This hybrid procedure eliminates the need for nested electronic and geometric optimization cycles found in classical Born-Oppenheimer approaches, enabling direct geometry relaxation within a single variational loop. Delgado *et al.* demonstrated that this approach achieves chemical accuracy² for small molecules (for example H_2 , H_2O) with shallow circuits, making it suitable for current NISQ devices and marking an important step toward quantum-assisted molecular design.

²Chemical accuracy: 1 kcal/mol (0.0016 Hartree), the standard benchmark in quantum chemistry corresponding to 1.6×10^{-3} Hartree.

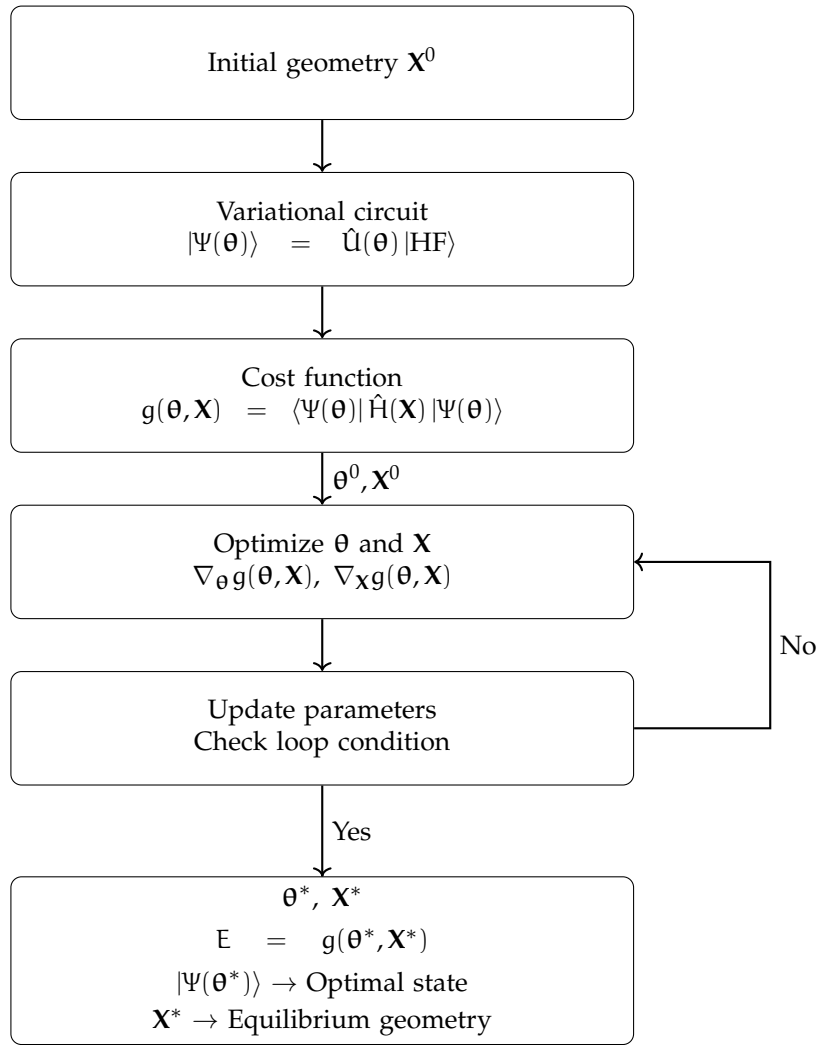


Figure 4.2: Workflow of the variational quantum algorithm for molecular geometry optimization. Circuit parameters $\boldsymbol{\theta}$ and nuclear coordinates $\mathbf{X}$ are jointly optimized until convergence in both total energy and geometry is achieved.

Chapter 5

Results

For all experiments, we used the PennyLane library for implementing the Variational Quantum Eigensolver (VQE) framework in Python. Classical electronic structure computations and molecular integrals were performed using the PySCF quantum chemistry package [46]. All calculations employed the minimal STO-3G basis set, which provides a compact yet consistent representation suitable for benchmarking quantum algorithms. The STO-3G basis is particularly suitable for small-scale simulations and for efficient mapping to qubit Hamiltonians in quantum chemistry.

All code implementations for the experiments described in this chapter are available at: https://github.com/Pinetreeez/Thesis_VQE_Molecular_Optimization.

The experiments were executed on classical simulators emulating ideal, noise-free quantum devices, allowing us to isolate and analyze the convergence properties of the hybrid variational optimization framework without the influence of hardware noise.

5.1 Molecular Geometry Optimization of H_3^+

To validate the proposed variational geometry optimization framework, we applied it to the H_3^+ molecular ion. Despite its small size, H_3^+ exhibits a non-trivial potential energy surface (PES), therefore it is an ideal test case for verifying the consistency between classical reference methods and quantum variational optimization. In the experiments, we compared our VQE results to the classical method FCI seen in 2.6.2.

In both experiments, the molecular Hamiltonian $\hat{H}(\mathbf{X})$ is parameterized by the nuclear coordinates $\mathbf{X}$ and expressed in the second-quantized form. The total energy expectation value is variationally minimized with respect to both the circuit parameters θ and the nuclear coordinates:

$$E(\theta, \mathbf{X}) = \langle \Psi(\theta) | \hat{H}(\mathbf{X}) | \Psi(\theta) \rangle.$$

Optimization proceeds until self-consistent convergence in both θ and $\mathbf{X}$ is achieved, corresponding to equilibrium geometry and ground-state energy.

5.1.1 H_3^+ with Hartree-Fock Initialization

Geometric Initialization

The molecular geometry of H_3^+ is initialized as an equilateral triangle with a side length of 0.88 Å. Placing the first hydrogen atom at the origin, the coordinates of the remaining two atoms can be determined geometrically using the trigonometric relations:

$$x = r \cos(60^\circ), \quad y = r \sin(60^\circ),$$

where $r = 0.88 \text{ \AA}$ is considered. This defines the primitive structure used as the initial nuclear configuration $\mathbf{X}^0$.

The following Python script shows the initialization and optimization workflow of the HF energy and optimized geometry:

Listing 5.1: HF-based geometry optimization for H_3^+ using PySCF.

```
from pyscf import gto, scf
from pyscf.geomopt import geometric_solver

# Define initial geometry (equilateral triangle, r = 0.88 A)
mol = gto.M(
    atom='''
    H  0.000    0.000    0.0
    H  0.000    0.880    0.0
    H  0.440    0.762    0.0 # x = 0.88*cos(60), y = 0.88*sin(60)
    ''',
    basis='sto-3g',
    charge=1, # H3+ has +1 charge
    unit='angstrom'
)

# Perform Hartree-Fock calculation
mf = scf.RHF(mol).run()
print("Initial HF energy:", mf.e_tot)

# Optimize geometry
mol_optimized = geometric_solver.optimize(mf)
print("Optimized coordinates (Bohr):", mol_optimized.atom_coords())

# Re-run HF on optimized geometry
mf_optimized = scf.RHF(mol_optimized).run()
print("Optimized HF energy:", mf_optimized.e_tot)
```

The Variational Quantum Algorithm

Next, we initialize the **Variational Quantum Eigensolver (VQE)** procedure using the optimized Hartree-Fock geometry obtained in the previous step. At this stage, the molecular Hamiltonian $\hat{H}(\mathbf{X}_{\text{HF}})$ is constructed from the one- and two-electron integrals corresponding to the HF-optimized coordinates. The electronic structure of the H_3^+ ion consists of **two electrons** distributed among **six spin orbitals**, which in turn define the number of qubits required for the quantum simulation (one qubit per spin orbital). Hence, the system is encoded in a six-qubit register, initially representing the Hartree-Fock reference configuration

$$|\Psi_{\text{HF}}\rangle = |110000\rangle,$$

where the first two spin orbitals are occupied and the remaining four are unoccupied.

To introduce electronic correlations beyond the mean-field approximation, we construct a **parameterized quantum circuit (ansatz)** using *double excitation operators*. These operations promote pairs of electrons from occupied to virtual orbitals, allowing the variational circuit to explore correlated configurations. Specifically, two double excitation gates are applied:

```
qml.DoubleExcitation(params[0], wires=[0, 1, 2, 3]),
qml.DoubleExcitation(params[1], wires=[0, 1, 4, 5]),
```

where each gate corresponds to an excitation from the occupied orbitals (0,1) to the virtual orbitals (2,3) and (4,5), respectively.

Following the circuit construction, the optimization loop jointly updates both the variational parameters θ and the nuclear coordinates $\mathbf{X}$. The cost function is defined as

$$E(\theta, \mathbf{X}) = \langle \Psi(\theta) | \hat{H}(\mathbf{X}) | \Psi(\theta) \rangle,$$

where $H(\mathbf{x})$ constructs the molecular Hamiltonian with geometry corresponding to positions $\mathbf{X}$ and the circuit returns its expectation value.

To optimize the nuclear geometry, the gradient with respect to the coordinates $\mathbf{X}$ is estimated using a **central finite-difference method**:

$$\frac{\partial f}{\partial x_i} \approx \frac{f(x_i + \frac{\delta}{2}) - f(x_i - \frac{\delta}{2})}{\delta},$$

implemented in the function `finite_diff`. This allows the algorithm to approximate $\nabla_{\mathbf{X}} g(\theta, \mathbf{X})$ without requiring analytic derivatives of the Hamiltonian.

The function `grad_x` combines this finite-difference evaluation with the circuit expectation values to obtain the geometry gradient. The optimization then proceeds iteratively: for each step, the parameter gradient $\nabla_{\theta} g$ and geometry gradient $\nabla_{\mathbf{X}} g$ are computed using automatic differentiation (`jax.grad`) and finite differences, respectively. The η_{θ} and $\eta_{\mathbf{X}}$ variables denote the learning rates for the parameter and geometry updates, respectively. These values determine the step size along each gradient direction, too small of a value can slow convergence, while too large of a value can lead to numerical instability or oscillations. In our implementation, both learning rates were empirically set to $\eta_{\theta} = \eta_{\mathbf{X}} = 0.8$:

$$\begin{aligned}\theta &\leftarrow \theta - \eta_{\theta} \nabla_{\theta} g, \\ \mathbf{X} &\leftarrow \mathbf{X} - \eta_{\mathbf{X}} \nabla_{\mathbf{X}} g.\end{aligned}$$

During the optimization, the total energy (in Hartree) and bond length (converted from Bohr to Ångstrom using the factor $1 \text{ Bohr} = 0.529177210903 \text{ Å}$) are recorded at each iteration:

$$d_{\text{H-H}} = \|\mathbf{X}_1 - \mathbf{X}_2\| \times 0.529177210903.$$

This enables monitoring of the convergence behavior of both the energy and the molecular geometry. The optimization loop terminates once the maximum component of the nuclear gradient satisfies

$$\max \|\nabla_{\mathbf{X}} g(\theta, \mathbf{X})\| \leq 10^{-4},$$

indicating that the system has reached equilibrium. The threshold value of 10^{-4} (in atomic units) is a standard convergence criterion in quantum chemistry in which the residual forces acting on each nucleus are negligible. At this point, further optimization would not lead to a significant change in either the total energy or the molecular geometry, signifying that the system has reached its minimum-energy configuration within numerical precision.

Finally, the code outputs the **optimized ground-state energy** E^* and the corresponding **equilibrium geometry**, printed in Cartesian coordinates:

$$\begin{aligned}E^* &= g(\theta^*, \mathbf{X}^*), \\ \mathbf{X}^* &= \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}.\end{aligned}$$

These results represent the converged configuration of the H_3^+ molecule obtained through the hybrid variational optimization procedure.

The parameters $\theta = \{\theta_0, \theta_1\}$ are optimized during the VQE loop to minimize the expectation value of the Hamiltonian:

$$E(\theta) = \langle \Psi(\theta) | \hat{H}(\mathbf{X}_{\text{HF}}) | \Psi(\theta) \rangle.$$

This variational procedure allows the circuit to capture electron correlation effects while remaining compatible with near-term quantum hardware constraints.

Figure 5.1 shows the convergence behavior of the variational quantum algorithm during the molecular geometry optimization of the H_3^+ ion, initialized with the Hartree-Fock reference state. The left panel depicts the difference between the VQE and FCI energies ($E_{\text{VQE}} - E_{\text{FCI}}$) as a function of the optimization step. The energy decreases monotonically and approaches chemical accuracy (on the order of 10^{-3} Hartree), demonstrating effective convergence of the variational parameters. The right panel illustrates the evolution of the bond length during optimization. The geometry stabilizes after a few iterations, converging toward an equilibrium bond length of approximately 0.986 Å, in agreement with the theoretical reference value for H_3^+ . These results confirm the successful coupling between quantum circuit optimization and nuclear coordinate relaxation within the hybrid variational framework.

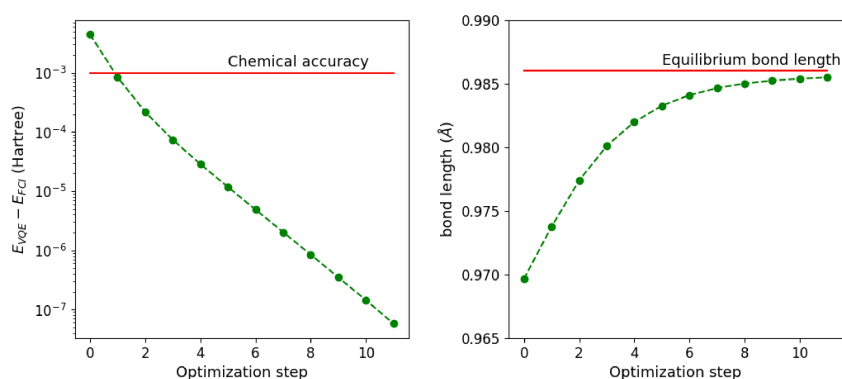


Figure 5.1: The results for H_3^+ initialized with the Hartree-Fock reference: **Left:** The difference between the VQE and FCI energies ($E_{\text{VQE}} - E_{\text{FCI}}$) as a function of the optimization step. **Right:** The evolution of the bond length during optimization.

5.1.2 H_3^+ with DFT Initialization

The second experiment employs Density Functional Theory (DFT) (described in Subsection 2.6.3) to initialize the electronic structure. The same initial geometry as in 5.1.1 is used. The DFT reference state and energy are computed using the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional within the restricted Kohn-Sham (RKS) formalism for closed-shell systems.

The corresponding initialization code is shown in Listing 5.2. After computing the initial self-consistent field (SCF) solution, the molecular geometry is optimized using the built-in `geometric_solver`, which iteratively relaxes atomic positions based on the DFT-calculated forces until convergence.

Listing 5.2: DFT-based geometry optimization for H_3^+ using PySCF and the PBE functional.

```
from pyscf import gto, dft
from pyscf.geomopt import geometric_solver

# Define initial geometry for H3+ (equilateral triangle)
mol = gto.M(
    atom=''''
    H 0.000 0.000 0.0
    H 0.000 0.880 0.0
    H 0.440 0.762 0.0 # x = 0.88*cos(60), y = 0.88*sin(60)
    ''',
    basis='sto-3g',
    charge=1,
```

```

        unit='angstrom',
        spin=0
    )

    # Initialize DFT with the PBE functional
    mf = dft.RKS(mol)
    mf.xc = 'PBE'
    mf.kernel()
    print("Initial DFT (PBE) energy:", mf.e_tot)

    # Optimize geometry
    mol_optimized = geometric_solver.optimize(mf)
    print("Optimized coordinates (Bohr):", mol_optimized.atom_coords())

    # Re-run DFT on optimized geometry
    mf_optimized = dft.RKS(mol_optimized)
    mf_optimized.xc = 'PBE'
    mf_optimized.kernel()
    print("Optimized DFT energy:", mf_optimized.e_tot)

```

This initialization is followed by the same variational quantum algorithm used in 5.1.1.

This algorithm yields a bond length of approximately 0.985 Å, consistent with literature values for the equilibrium geometry of H_3^+ . This optimized structure serves as the starting point for the hybrid variational optimization. Because DFT provides a more accurate description of electron correlation at the classical level, the quantum optimization starts closer to the minimum of the potential energy surface, resulting in faster convergence.

Figure 5.2 illustrates the convergence of the variationally obtained energy and bond length with respect to the optimization step. The left panel shows the logarithmic decrease of the energy deviation ($E_{\text{VQE}} - E_{\text{FCI}}$), achieving chemical accuracy (10^{-3} Hartree) after a single iteration. The right panel shows the corresponding stabilization of the bond length around 0.985 Å, confirming that the equilibrium configuration is reached efficiently.

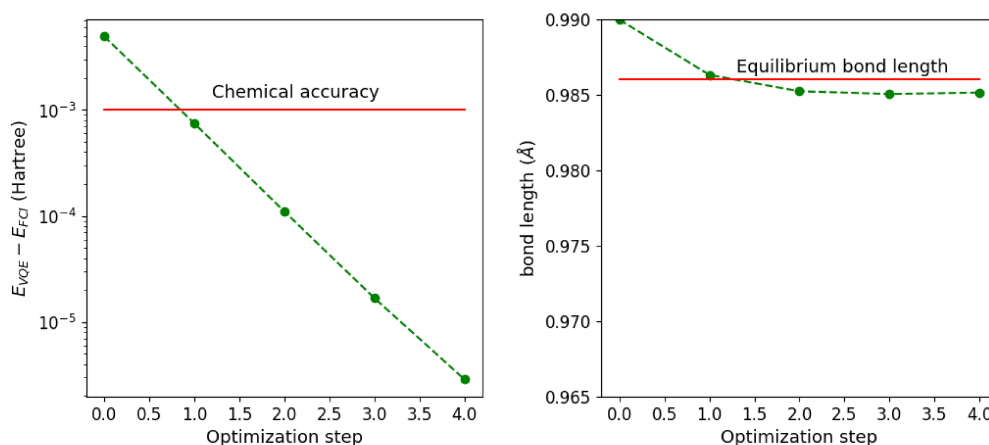


Figure 5.2: Convergence behavior of H_3^+ initialized with the DFT (PBE) reference. **Left:** Energy deviation from the FCI reference decreases exponentially, reaching chemical accuracy after the first iteration. **Right:** The optimized bond length stabilizes at 0.985 Å after four iterations, in agreement with the equilibrium value predicted by DFT.

5.1.3 Comparison and Discussion

Table 5.1 summarizes the performance of the two initialization strategies. Both the Hartree-Fock and DFT initializations ultimately converge to nearly identical equilibrium bond

lengths and ground-state energies. However, the DFT-based initialization reaches the same accuracy in fewer optimization steps due to its more accurate initial electronic density and geometry.

Table 5.1: Comparison of optimized results for H_3^+ using different initialization strategies.

Initialization	Energy Deviation from FCI (Hartree)	Equilibrium Bond Length ($\text{\AA}$)	Steps Until Convergence
Hartree-Fock (HF)	1.2×10^{-4}	0.986	11
DFT (PBE)	4.5×10^{-5}	0.985	4

These results confirm that the variational optimization framework successfully reproduces the equilibrium geometry and energy of H_3^+ within chemical accuracy. Furthermore, initializing the optimization from a DFT-derived geometry improves convergence speed and stability, highlighting the benefit of combining classical DFT methods with quantum variational algorithms for molecular structure prediction.

5.2 Molecular Geometry Optimization of H_2

To further validate the variational geometry optimization framework, we applied the same procedure to the dihydrogen molecule (H_2). This system represents the simplest molecular case, consisting of two protons and two electrons.

As in the previous section, both Hartree-Fock (HF) and Density Functional Theory (DFT) initializations were tested using the minimal STO-3G basis set implemented in PySCF. The DFT calculation employed the PBE exchange-correlation functional, and both methods started from an initial H-H bond length of $0.725 \text{ \AA}$. After performing variational optimization using the same variational quantum algorithm loop as described in Section 5.1, both initialization schemes converged to the same equilibrium geometry and ground-state energy within five optimization steps. Therefore, only the DFT-based result is presented here for clarity.

Figure 5.3 illustrates the convergence behavior of the VQE algorithm applied to H_2 . The left panel shows the logarithmic decrease of the energy deviation from the Full Configuration Interaction (FCI) reference, $(E_{\text{VQE}} - E_{\text{FCI}})$, as a function of the optimization step. The VQE energy rapidly reaches the chemical accuracy threshold (1.6 mHa) within the first iteration, confirming the efficiency of the variational approach for this two-electron system. The right panel depicts the corresponding evolution of the H-H bond length, which converges smoothly towards the equilibrium value of approximately $0.735 \text{ \AA}$ in five optimization steps, in agreement with the STO-3G reference geometry.

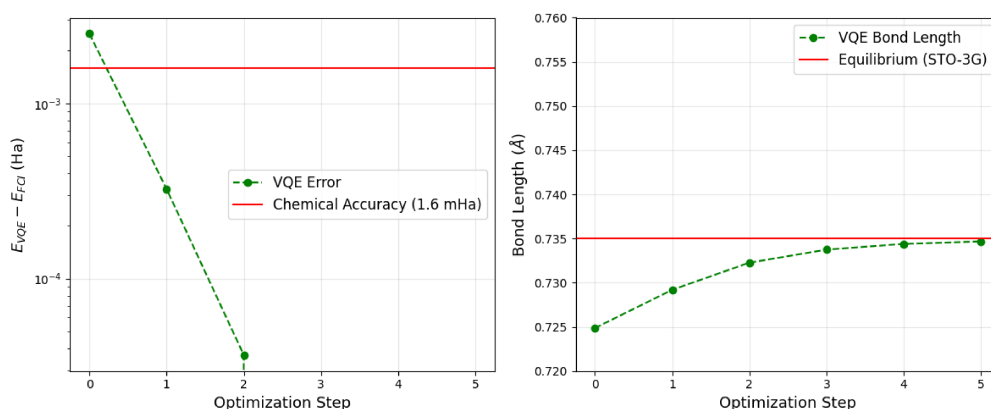


Figure 5.3: Convergence behavior of the VQE-based geometry optimization for H_2 . **Left:** Energy deviation from the FCI reference decreases below the chemical accuracy threshold within a single iteration. **Right:** Bond length evolution during optimization, converging to the equilibrium distance (0.735 Å) predicted by the STO-3G reference.

Both initialization strategies (HF and DFT) produced identical results for H_2 , indicating that in this simple system, the initial electronic structure has negligible influence on convergence. This outcome is expected given the simplicity of the H_2 molecule. Nevertheless, these results further confirm the accuracy of the proposed variational optimization framework when applied to small molecular systems.

Chapter 6

Conclusions

6.1 Summary and Discussion

In this thesis, we developed and implemented a hybrid quantum-classical framework for **molecular geometry optimization** based on the **Variational Quantum Eigensolver (VQE)**. Our approach combines classical electronic-structure initialization (using both Hartree-Fock and Density Functional Theory) with quantum variational optimization to determine equilibrium geometries and ground-state energies of molecular systems.

Through systematic evaluation on benchmark systems (H_2 and H_3^+), we demonstrated that the proposed framework achieves convergence within **chemical accuracy** ($\sim 10^{-3}$ Ha) and reproduces equilibrium bond lengths consistent with Full Configuration Interaction (FCI) reference data.

For the H_2 molecule, both initialization strategies produced identical results, which is expected for such a simple two-electron system where electronic correlation effects are minimal.

However, the more complex H_3^+ system, when initialized with DFT-derived electronic structures, exhibited faster convergence compared to Hartree-Fock initialization, confirming our hypothesis that DFT provides a more accurate starting point for variational optimization.

6.2 Computational Limits

While VQE provides a conceptually powerful method for molecular optimization, its practical execution on classical simulators remains computationally demanding. The simulation cost scales exponentially with the number of qubits, limiting classical reproduction to only a few-atom systems. For example, attempts to simulate the water molecule (H_2O) using the same workflow requires several days of computation on a standard computer. This highlights the necessity of high-performance computing resources or access to real quantum processors for larger systems.

Moreover, as system size increases, the design of efficient ansatz and the mitigation of quantum noise become critical to maintaining accuracy and scalability. Thus, while the presented results validate the framework conceptually, further optimization of circuit depth and parameterization strategies is essential for extending applicability to more complex molecules.

6.3 Future Work

Future research could extend this work in several directions:

- **Scaling to larger molecules:** Implementing the framework on high-performance computing platforms or real quantum hardware to study polyatomic systems such as H_2O or CH_4 . In the paper [21], the algorithm was successfully executed for H_2O (without including the DFT-based initialization proposed in this work), but in our case, running the same experiment on a local machine was computationally infeasible. This shows that while the framework is scalable in principle, practical execution for larger molecules remains computationally demanding without access to high-performance or quantum hardware.
- **Ansatz optimization:** Exploring adaptive or problem-tailored ansatz (for example, ADAPT-VQE) to improve convergence and reduce circuit depth [47].
- **Error mitigation and noise handling:** Incorporating measurement-error mitigation and readout correction techniques to improve experimental accuracy on NISQ devices.
- **Applications beyond molecular geometry:** Extending the VQE framework to other domains, such as computing excited-state energies, simulating reaction pathways, or studying material properties like band gaps and magnetic interactions. These applications would further demonstrate the versatility of variational quantum algorithms in quantum chemistry and condensed matter physics [13].

6.4 Concluding Remarks

This thesis demonstrates that integrating **DFT-based initialization** with **variational quantum algorithms** provides a promising route for accelerating molecular geometry optimization. The results confirm that hybrid quantum-classical workflows can already reproduce chemically accurate results for small molecules, even on simulated hardware, laying the foundation for future large-scale quantum chemistry applications as quantum devices continue to mature.

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